Combinatorial Geometry

Labix

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Contents

1 Simplicies

1.1 Basic Definitions

Definition 1.1.1 (Affinely Independent)
Let $n,m \in \mathbb{N}$. We say that a set of points $\{v_0,\dots,v_m\} \subset \mathbb{R}^n$ is affinely independent if $v_1 - v_0,\dots,v_n - v_0$ are linearly independent.

Definition 1.1.2 (n-Simplexes)
Let $n \in \mathbb{N}$. Let $v_0,\dots,v_n \in \mathbb{R}^n$ be a set of affinely independent points. An n-simplex is the set of points

$$\Delta^n = \left\{\sum_{k=0}^n t_k v_k \,\middle|\, \sum_{k=0}^n t_k = 1 \text{ and } t_k \geq 0 \text{ for all } k = 0,\dots,n\right\}$$

We write $\Delta^n = [v_0,\dots,v_n]$ to indicate the spanning vectors. The standard n-simplex is just the n-simplex whose vertices are the standard basis vectors for $\mathbb{R}^{n+1}$.

Note that the vertices here are an ordered set so that an orientation is inherently defined. Realistically, the order of the vertices does not change the homology groups which we will define later.

Definition 1.1.3 (Properties of n-Simplexes) Let $\Delta^n = [v_0,\dots,v_n]$ be an n-simplex.
- The kth face of Δ^n is a the $n-1$ simplex $\Delta^n_k = \Delta^n \cap \{x_k = 0\}$. We use $[v_0,\dots,\hat{v}_k,\cdots,v_n]$ to indicate the kth $n-1$dimensional face
- The boundary of Δ, written $\partial\Delta^n$ is the union of all its proper faces
- The interior is defined to be $(\Delta^n)^\circ = \Delta^n \setminus \partial\Delta^n$

It is easy to see that any face of an n-simplex is an $n-1$ simplex in its own right.

Lemma 1.1.4 Any two k-simplexes where one in $\mathbb{R}^m$ and one in $\mathbb{R}^n$ are homeomorphic.

1.2 Barycentric Subdivision

Barycentric subdivisions is the main ingredient for proving one of the major results of singular homology. It is defined first through Δ-set, and then passed on to singular n-simplexes, which we will see in the next chapter.

Definition 1.2.1 (Barycenter) Let $v = [v_0,\dots,v_n]$ be an n-simplex. Define the barycenter of v to be the point

$$B(v) = \frac{1}{n+1}\sum_{i=0}^n v_i$$

In fact, we can find the barycenter inductively. Knowing the barycenter b_i on the ith face $[v_0,\dots,\hat{v}_i,\dots,v_n]$, let l_i be the line connecting b_i and v_i. Then b is the intersection of all these lines l_i.

Definition 1.2.2 (Barycentric Cone) Let v be an n-simplex. Let $w = [w_1,\dots,w_n] \subseteq v$ be an $(n-1)$-simplex in v. Define the barycentric cone of w to be the n-simplex

$$\mathcal{B}(w) = [B(v), w_1,\dots,w_n] \subset v$$

Intuitively, this means that given a face of Δ^n which is an $(n-1)$-simplex, its barycentric cone is the n-simplex constructed from the vertices of the face together with the barycenter.

Definition 1.2.3 (The Barycentric Map) Let v be an n-simplex. Define the barycentric map of v to be the map

$$\mathcal{B} : C_{n-1}(v) \to C_n(v)$$

defined on the basis by $\mathcal{B}([w_1, \dots, w_n]) = [B(v), w_1, \dots, w_n]$.

Definition 1.2.4 (Barycentric Subdivision) Let Δ^n be the standard n-simplex. Define inductively the barycentric subdivision

$$S^\Delta(\Delta^n) \in C_n(\Delta^n)$$

of Δ^n by the following recursive formula.

- When $n = 0$, $S^\Delta(\Delta^0) = \Delta^0$
- When $n > 0$, the barycentric subdivision of Δ^n is given by

$$S^\Delta(\Delta^n) = \sum_{i=0}^{n} (-1)^i \mathcal{B} S^\Delta(\partial_i \Delta^n)$$

To elicit a few examples, consider the case of $n = 1$. Then we have

$$\begin{aligned} S(\Delta^1) &= \mathcal{B}S(\partial\Delta^1) \\ &= \mathcal{B}S(\partial_0\Delta^1) - \mathcal{B}S(\partial_1\Delta^1) \\ &= \mathcal{B}[v_1] - \mathcal{B}[v_0] \\ &= [b, v_1] - [b, v_0] \end{aligned}$$

where $b = \frac{1}{2}(v_0 + v_1)$. Intuitively, we are breaking up the n-simplex Δ^n into tiny pieces of n-simplexes using the center of mass of each subsimplex in Δ^n.

Lemma 1.2.5 Let $\sigma = [w_1, \dots, w_n] \subseteq \Delta^n$ be an $(n-1)$-simplex. Then the following are true.

- $\partial(\mathcal{B}(\sigma)) + \mathcal{B}(\partial\sigma) = \sigma$
- $\partial S(\Delta^n) = S(\partial\Delta^n)$

Proof We have that

$$\begin{aligned} \partial\mathcal{B}[w_1, \dots, w_n] &= \partial[b, w_1, \dots, w_n] \\ &= \sum_{i=0}^{n} (-1)^i \partial_i [b, w_1, \dots, w_n] \\ &= [w_1, \dots, w_n] - \mathcal{B}(\partial[w_1, \dots, w_n]) \end{aligned}$$

And thus the first identity is satisfied.
We prove the second item inductively. When $n = 0$, we have 0 on both sides. When $n > 0$, we have

$$\begin{aligned} \partial S\Delta^n &= \partial\mathcal{B}(S(\partial\Delta^n)) \\ &= \mathrm{id}(S\partial\Delta^n) - \mathcal{B}(\partial(S\partial\Delta^n)) && \text{(First Identity)} \\ &= S\partial\Delta^n - \mathcal{B}(S\partial^2\Delta^n) && \text{(Induction)} \\ &= S\partial\Delta^n \end{aligned}$$

■

Lemma 1.2.6 Let $v = [v_0, \dots, v_n]$ be a simplex. Let $[w_0, \dots, w_n]$ be a component in the barycentric subdivision of v. Then we have

$$\mathrm{diam}([w_0, \dots, w_n]) \leq \frac{n}{n+1} \mathrm{diam}([v_0, \dots, v_n])$$

Proof If $n = 0$, then the claim is true since $[w_0] = [v_0]$ has diameter 0. Assume that $n > 0$. We first prove the following subclaim: For every $x \in [x_0, \dots, x_n]$ an n-simplex, its maximum distance to points in the simplex is attained at a vertex x_i. Indeed if $y \in [x_0, \dots, x_n]$ with $\|x - y\|$ maximal, then we can write $y = \sum_{i=1}^n t_i x_i$ with $\sum_{i=1}^n t_i = 1$ and $t_i \geq 0$. Then we have that

$$\begin{aligned} \|x - y\| &= \left\| x - \sum_{i=1}^n t_i x_i \right\| \\ &= \left\| \sum_{i=1}^n t_i (x - x_i) \right\| \\ &\leq \sum_{i=1}^n t_i \|x - x_i\| \\ &\leq \max \|x - x - i\| \end{aligned}$$

with equality if y is one of the vertices with $\|x - x_i\|$ being maximal. By applying the above twice, we have that the diameter of $[w_0, \dots, w_n]$ is the length of the longest edge $[x, y]$ in $[w_0, \dots, w_n]$. Now there are two cases.

Case 1: None of x, y is the barycenter b of $[w_0, \dots, w_n]$. Then they must be the vertices of a simplex in the barycentric subdivision of one of the faces $[v_0, \dots, \hat{v_i}, \dots, v_n]$. By induction, we have that

$$\begin{aligned} \text{diam}([w_0, \dots, w_n]) &= \|x - y\| \\ &\leq \frac{n-1}{n} \text{diam}([v_0, \dots, \hat{v_i}, \dots, v_n]) \\ &\leq \frac{n}{n+1} \text{diam}([v_0, \dots, v_n]) \end{aligned}$$

and so we are done.

Case 2: Without loss of generality, $x = b$.
Then y lies on some face of $[v_0, \dots, v_n]$ and the claim above implies that we can take $y = v_i$ for some vertex v_i of that face. Let b_i be the barycenter of $[v_0, \dots, \hat{v_i}, \dots, v_n]$. Then we have that the barycenter of $[v_0, \dots, v_n]$ is

$$b = \frac{1}{n+1} \sum_{j=1}^n v_j = \frac{1}{n+1} v_i + \frac{n}{n+1} b_i$$

We thus have that

$$\begin{aligned} \text{diam}([w_0, \dots, w_n]) &= \|v_i - b\| \\ &\leq \frac{n}{n+1} \|v_i - b_i\| \\ &\leq \frac{n}{n+1} \text{diam}([v_0, \dots, v_n]) \end{aligned}$$

and so we conclude. ■

2 Simplicial and Delta Complexes

2.1 Delta Complexes

Definition 2.1.1 (Δ-Complexes)
A Δ-complex $(S_\bullet, d_\bullet)$ consists of the following data.

- For each $n \in \mathbb{N}$, a set S_n of n-simplicies.
- For each $n \in \mathbb{N}$ and $0 \leq i \leq n$, a map $d_i^n : S_n \to S_{n-1}$ such that

$$d_i^{n-1} \circ d_j^n = d_{j-1}^{n-1} \circ d_i^n$$

called the face relation is satisfied whenever $i < j$.

In particular, d_i sends an n-simplex to its ith face, which is an $n-1$ simplex. This means that for $s = [v_1, \dots, v_n] \in S_n$, $d_i(s) = [v_1, \dots, \hat{v}_i, \cdots, v_n]$.

Definition 2.1.2 (Geometric Realization of Δ-Complexes)
Let $S = (S_\bullet, d_\bullet)$ be a Δ-set. The geometric realization of S is a topological space X that is built up inductively as follows:

- The 0-skeleton X^0 is a discrete set with points in S_0
- Given the $n-1$ skeleton X^{n-1} and S_{n-1}. Now define

$$X^n = \left(X^{n-1} \cup \coprod_{\alpha \in I_n} \Delta_\alpha^n \right) / \sim$$

where $\sim$ is the equivalence relation $\Delta_\beta^{n-1} \sim \partial_k \Delta_\alpha^n$ given from the face maps $d_\bullet$. (Intuitively, each face of the n-simplex in S_n gets identified with a $n-1$ simplex in X^{n-1})
- Define $X = \bigcup_n X^n$. The minimal n for which $X = X^n$ is called the dimension of n

We also write X as $|S|$ to indicate the Δ-set.

Definition 2.1.3 (Map of Delta-Complexes) Let $S = (S_\bullet, d_\bullet)$ and $S' = (S'_\bullet, d'_\bullet)$ be two Δ-sets. A map of Δ-sets $f : S \to S'$ is a family of maps $f_n : S_n \to S'_n$ for each $n \in \mathbb{N}$ such that for all $0 \leq i \leq n$, we have that

$$d'_i \circ f_n = f_{n-1} \circ d_i : S_n \to S'_{n-1}$$

2.2 Simplicial Complexes

Definition 2.2.1 (Simplicial Complexes) Let $\mathcal{K}$ be a set of simplexes. We say that $\mathcal{K}$ is a simplicial complex if the following are true.

- For any $S \in \mathcal{K}$, every face $\partial_k S$ of S lies in $\mathcal{K}$.
- If S_1 and S_2 are two simplexes in $\mathcal{K}$, then either $S_1 \cap S_2$ is empty or $S_1 \cap S_2$ is a common face of both S_1 and S_2.

Theorem 2.2.2 Every simplicial complex is a Δ-complex.

Definition 2.2.3 (Simplicial Maps)
Let $\mathcal{K}, m\mathcal{L}$ be simplicial complexes. Let $f : \mathcal{K} \to \mathcal{L}$ be a function. We say that f is a simplicial map if the following are true.

- There exists a simplex $T \in \mathcal{L}$ such that $f(S) \subseteq T$.
- For all $S \in \mathcal{K}$, $f|_S$ is affine linear.

Definition 2.2.4 (Piecewise Linear Maps)
Let $\mathcal{K}, m\mathcal{L}$ be simplicial complexes. Let $f : \mathcal{K} \to \mathcal{L}$ be a function. We say that f is piecewise linear if there is a subcomplex $\mathcal{K}' \subseteq \mathcal{K}$ such that $f|_{\mathcal{K}'}$ is a simplicial map.

2.3 Abstract Simplicial Complexes

Definition 2.3.1 (Abstract Simplicial Complex)
Let T be a set. Let $S \subseteq \mathcal{P}(T)$. We say that S is an abstract simplicial complex if for all $U \in S$ and $V \subseteq U$, we have $V \in S$.

We mostly consider the case where T is finite. In this case, we may replace the underlying set T by $\{1, \dots, n\}$ by a suitable $n \in \mathbb{N}$.

Example 2.3.2
The set $S = \mathcal{P}(\{1, 2, 3\}) \cup \mathcal{P}(\{1, 3, 4\}) \cup \mathcal{P}(\{2, 4\})$ is an abstract simplicial complex.

Definition 2.3.3 (Geometric Realization of Abstract Simplicial Complexes)
Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Define the geometric realization of S to be the space

$$|S| = \bigcup_{I \in S} \Delta^I$$

where Δ^I is the $|I|$-simplex with vertices e_i for $i \in I$.

By definition, the geometric realization $|S|$ is a subcomplex of the standard n-simplex $[e_1, \dots, e_{n+1}] \subseteq \mathbb{R}^{n+1}$.

Example 2.3.4
Let $n \in \mathbb{N}$. Consider the set $S = \mathcal{P}(\{1, \dots, n+1\})$. Then $|S|$ is the standard n-simplex.

2.4 The Face Vector of a Simplicial Complex

Definition 2.4.1 (The Face Vector)
Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Define the face vector $(f_{-1}, \dots, f_n)$ of S by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } S$$

for $0 \leq i \leq n$ and by convention, $f_{-1} = 1$.

Definition 2.4.2 (The h-Vector)
Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Let $(f_0, \dots, f_n)$ be the face vector of S. Define the h-vector $(h_0, \dots, h_n)$ of S by the formula

$$h_i = \sum_{k=0}^{i} (-1)^{i-k} \binom{n+1-k}{i-k} f_{k-1}$$

Lemma 2.4.3
Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Let $(f_0, \dots, f_n)$ be the face vector of S. Let $(h_0, \dots, h_n)$ be the h-vector of S. Then the following are true.

- We have
$$\sum_{k=0}^{n+1} f_{k-1}(t-1)^{d-k} = \sum_{j=0}^{n+1} h_j t^{n+1-j}$$
for variable t.
- For $0 \leq i \leq n$, we have
$$f_i = \sum_{k=0}^{i+1} \binom{n+1}{i+1-k} h_k$$

2.5 Cubical Embeddings of Simplicial Complexes

Definition 2.5.1 (The Cubical Embedding)
Let $n \in \mathbb{N}$. Define the cubical embedding $i_c : \Delta^n \to [0,1]^{n+1}$ as follows.

1. For $1 \leq k \leq n$, the barycenter of the k-dimensional face $B([e_{i_1}, \dots, e_{i_k}])$ is sent to $(u_1, \dots, u_{n+1}) \in [0,1]^{n+1}$ where $u_j = 0$ if $j \in \{i_1, \dots, i_k\}$ and $u_j = 1$ otherwise.
2. Extend this into a piecewise linear embedding $i_c : \Delta^n \to [0,1]^{n+1}$ as follows. For each simplex $[p_1, \dots, p_l]$ in the barycentric subdivision of Δ^n, $[i_c(p_1), \dots, i_c(p_l)]$ lies on a face of $[0,1]^{n+1}$. Hence there exists an simplicial map sending $[p_1, \dots, p_l]$ to $[i_c(p_1), \dots, i_c(p_l)]$.

Lemma 2.5.2
Let $n \in \mathbb{N}$. Then $\partial[0,1]^n$ is a simplicial complex.

Definition 2.5.3 (Cubical Embeddings of Abstract Simplicial Complexes)
Let $\mathcal{K}$ be an abstract simplicial complex. Define the cubical embedding of $\mathcal{K}$ to be the subspace

$$\text{cc}(\mathcal{K}) = i_c(|\mathcal{K}|)$$

3 Polyhedrons and Polytopes

3.1 Convex Geometry

Definition 3.1.1 (Convex Sets)
Let $C \subseteq \mathbb{R}^n$. We say that C is convex if for all $x, y \in C$ and $t \in [0, 1]$, we have $(1-t)x + ty \in C$.

Definition 3.1.2 (Convex Hull)
Let $S \subseteq \mathbb{R}^n$ be a subset. Define the convex hull of S to be

$$\text{Conv}(S) = \left\{ \sum_{k=1}^{n} \lambda_k x_k \;\middle|\; x_k \in S \text{ and } \sum_{k=1}^{n} \lambda_k = 1 \right\}$$

Proposition 3.1.3
Let $S \subseteq \mathbb{R}^n$ be a subset. Then the following are sets are equal.

- The convex hull $\text{Conv}(S)$.
- $\bigcap_{C \subseteq \mathbb{R}^n \text{ is convex}} C$.
- The smallest convex set containing S.

3.2 Polyhedrons and its Properties

Polytopes exists more generically in Euclidean space. We lose nothing discussing polytopes in $\mathbb{R}^n$ since Euclidean spaces maybe identified with $\mathbb{R}^n$ canonically.

Definition 3.2.1 (Halfspaces)
A halfspace of $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ of the form

$$\{x \in \mathbb{R}^n \mid \langle a, x \rangle + b \geq 0 \mid \}$$

for some $a, b \in \mathbb{R}^n$.

Definition 3.2.2 (Polyhedrons)
A polyhedron in $\mathbb{R}^n$ is an intersection of finitely many halfspaces.

Definition 3.2.3 (Polytopes)
A polytope in $\mathbb{R}^n$ is a polyhedron that is bounded.

Proposition 3.2.4
Let $P \subseteq \mathbb{R}^n$ be a subset. Then P is a polytope if and only if there exists a finite subset $S \subseteq \mathbb{R}^n$ such that $P = \text{Conv}(S)$.

Definition 3.2.5 (Dimension)
Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the dimension of P to be the dimension of the smallest affine subspace containing P.

Definition 3.2.6 (Affinely Isomorphic)
Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are affinely isomorphic if there exists an affine transformation $f : \mathbb{R}^{n_1} \to \mathbb{R}^{n_2}$ such that $f|_{P_1}$ is a bijection.

Note that f itself need not be an affine isomorphism (i.e. an isomorphism of affine spaces).

3.3 Faces

Definition 3.3.1 (Supporting Hyperplane)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $H \subseteq \mathbb{R}^n$ be a hyperplane. We say that H is a supporting hyperplane for P if the following are true.

- $P \cap H \neq \emptyset$.
- P lies entirely in one of the halfspaces defined by H.

Definition 3.3.2 (Faces)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. A face of P is a subset of P defined by

$$F = P \cap H$$

where H is a supporting hyperplane for P.

Lemma 3.3.3
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face. Then F is a polytope.

Definition 3.3.4 (Vertices, Edges and Facets)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face.

- We say that F is a vertex if $\dim(F) = 0$.
- We say that F is an edge if $\dim(F) = 1$.
- We say that F is a facet if $\dim(F) = \dim(P) - 1$.

Definition 3.3.5 (The Face Poset)
Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Define the face poset $\mathcal{F}(P)$ of P to be the set of faces of P together with inclusion.

Definition 3.3.6 (The Face Vector)
Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d. Define the face vector $(f_0, \dots, f_{d-1})$ of S by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } P$$

for $0 \leq i \leq d$ and by convention, $f_{-1} = f_d = 1$.

Lemma 3.3.7 (Euler's Formula)
Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d. Then we have

$$\sum_{i=0}^{d-1} (-1)^i f_i = 1 + (-1)^d$$

Definition 3.3.8 (Combinatorially Equivalent)
Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are combinatorially equivalent if there exists a inclusion preserving bijection between $\mathcal{F}(P_1)$ and $\mathcal{F}(P_2)$. In this case we write $P_1 \simeq P_2$.

4 Cones and Fans

4.1 Cones and its Properties

Definition 4.1.1 (Cones)
Let $C \subseteq \mathbb{R}^n$. We say that C is a cone if $x \in C$ implies that $sx \in C$ for all $s \geq 0$.

Definition 4.1.2 (Polyhedral Cones)
A cone in $\mathbb{R}^n$ is a subset $C \subseteq \mathbb{R}^n$ such that if $c_1, \dots, c_n \in C$, then $\sum_{i=1}^n \lambda_i c_i \in C$ for all $\lambda_1, \dots, \lambda_n \geq 0$.

Definition 4.1.3 (Conical Hull)
Let $S \subseteq \mathbb{R}^n$ be finite. Define the conical hull of S to be

$$\text{Cone}(S) = \{\sum_{s\in S} \lambda_s s \mid \lambda_s \geq 0\}$$

Proposition 4.1.4
Let $C \subseteq \mathbb{R}^n$ be a subset. Then the following are equivalent.
- C is a polyhedral cone.
- There exists a finite subset $S \subseteq \mathbb{R}^n$ such that $C = \text{Cone}(S)$.
- C is the intersection of finitely many halfspaces containing 0 in their boundary.

Therefore cones are also polyhedrons.

Definition 4.1.5 (Dimension of a Cone)
Let $C \subseteq \mathbb{R}^n$ be a cone. Define the dimension of C to be the dimension of the smallest affine subspace containing C.

Definition 4.1.6 (Rational Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is rational if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that $C = \text{Cone}(S)$.

Definition 4.1.7 (Regular Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is regular if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that the following are true.
- $C = \text{Cone}(S)$.
- Elements of S are $\mathbb{Z}$-linearly independent.

4.2 The Dual Cone

Definition 4.2.1 (The Dual of a Cone)
Let $C \in \mathbb{R}^n$ be a cone. Define the dual of C to be

$$C^\vee = \{x \in \mathbb{R}^n \mid \langle x, c\rangle \geq 0 \text{ for all } c \in C\}$$

Definition 4.2.2 (Associated Hyperplane and Halfspace of a Linear Functional)
Let $\phi \in (\mathbb{R}^n)^*$. Define the associated hyperplane of ϕ to be

$$H_\phi = \{x \in \mathbb{R}^n \mid \phi(x) = 0\}$$

Define the associated halfspace of ϕ to be

$$H_\phi^+ = \{x \in \mathbb{R}^n \mid \phi(x) \geq 0\}$$

Lemma 4.2.3
Let $C \in \mathbb{R}^n$ be a cone. Let H be a hyperplane in $\mathbb{R}^n$. Then H is a supporting hyperplane of C if and only if there exists $\phi \in (\mathbb{R}^n)^*$ such that $H = H_\phi$.

Lemma 4.2.4
Let $C \in \mathbb{R}^n$ be a polyhedral cone. Then the following are true.

- $C^\vee$ is a polyhedral cone.
- $(C^\vee)^v ee = C$.
- $C = \bigcap_{i=1}^n H_{\phi_i}$ for $\phi_i \in (\mathbb{R}^n)^*$ if and only if $C^\vee = \text{Cone}(\phi_1, \dots, \phi_n)$.

Lemma 4.2.5 (Separation Lemma)
Let $C_1, C_2 \subseteq \mathbb{R}^n$ be cones such that $F = C_1 \cap C_2$ is a face of both C_1 and C_2. Then there exists $\phi \in C_1^\vee \cap (-C_2)^\vee$ such that

$$F = C_1 \cap H_\phi = C_2 \cap H_\phi$$

Definition 4.2.6 (Strongly Convex Cones)
Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is strongly convex if C does not contain a line.

Lemma 4.2.7
Let $C \subseteq \mathbb{R}^n$ be a cone. Then the following are equivalent.

- C is strongly convex.
- $\{0\}$ is a face of C.
- $C \cap (-C) = \{0\}$.
- $\dim(C^\vee) = n$.

4.3 Fans and its Properties

Definition 4.3.1 (Fans)

Definition 4.3.2 (Types of Fans)

Definition 4.3.3 (Complete Fans)

5 Polar Sets

5.1 Polar Sets

Definition 5.1.1 (Polar Sets)
Let $S \subseteq \mathbb{R}^n$ be a subset. Define the polar set of S to be

$$S^* = \{f \in (\mathbb{R}^n)^* \mid f(x) \leq 1 \text{ for all } x \in S\}$$

Proposition 5.1.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. If $0 \in P$, then P^* is a polytope.

Example 5.1.3
The polar set of the cube $C^3 = \{(a, b, c) \in \mathbb{R}^3 \mid -1 \leq a, b, c \leq 1\}$ is the octahedron

$$(C^3)^* = \{(x, y, z) \in (\mathbb{R}^3)^* \mid \pm x + \pm y + \pm z \leq 1\}$$

$P = (P^*)^*$.

$\mathcal{F}(P)^{\text{op}} \cong \mathcal{F}(P^*)$.

5.2 Simplicial Polytopes

Definition 5.2.1 (Simplicial Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simplicial polytope if all its facets are simplicies.

Lemma 5.2.2
Let $P \subseteq \mathbb{R}^n$ be a polytope. Then ∂P is a simplicial complex.

Proposition 5.2.3 (The Dehn-Sommerville Equation)
Let $P \subseteq \mathbb{R}^n$ be a simplicial polytope of dimension d. Then the following are true:

$$\sum_{j=k}^{d-1} (-1)^j \binom{j+1}{k+1} f_j = (-1)^{d-1} f_k$$

for $-1 \leq k \leq d-2$.

In particular, ∂P is both a polytope and a simplicial complex. The notion of face vector coincides, and they enjoy both the Dehn-Sommerville equation and its relation with the h-vector.

5.3 Simple and Simplicial Polytopes

Definition 5.3.1 (Simple Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simple polytope if for all vertices $v \in P$, there exists n edges $e_1, \dots, e_n \subseteq P$ such that $v \in \partial e_i$ for all i.

Proposition 5.3.2
Let $P \subseteq \mathbb{R}^n$. Then P is simple if and only P^* is simplicial.

Duality of d-dimensional faces with $(n-d)$-dimensional faces.

Proposition 5.3.3
Let $P \subseteq \mathbb{R}^n$ be a polytope. If P is both simple and simplicial, then P belongs to one of the following types of polytopes.

- $n = 2$ (and so P is a polygon).

- P is a simplex.

6 Some Results on the Classification of Polytopes

Let P be a simplicial polytope of dimension 4. Since P is homeomorphic to D^4, the following classification problems are equivalent.

- Classification of simplicial polytopes up to uniqueness of the face vector.
- Classification of triangulations of the 3-sphere up to uniqueness of the face vector of the simplicial complex.

(More subtle, this is not true for higher dimensions: see https://math.mit.edu/ rstan/pubs/pubfiles/54.pdf)

Of course, two polytopes that are not combinatorially equivalent can still have the same face vector. Let (f_0, f_1, f_2, f_3) be the face vector of P. A few is known in this particular case.

- Since P is simplicial, we have the bound $f_i \leq \binom{f_0}{i+1}$ from here https://en.wikipedia.org/wiki/Cyclic$_{p}olytope$
- The Euler characteristic of the sphere tells us that $f_0 - f_1 + f_2 - f_3 = 0$.
- https://arxiv.org/pdf/math/0604018 shows that $f_1 \geq 4f_0 - 10$.
- The same paper tells us that $f_2 = 2f_1 - 2f_0$ and $f_3 = f_1 - f_0$ (also this paper: https://math.mit.edu/ rstan/pubs/pubf
- For every pair of (f_0, f_1) such that $f_0 \geq 5$ and $\frac{f_0(f_0-1)}{2} \geq f_1 \geq 4f_0 - 10$, there exists a triangulation of the 3-sphere whose face vectors are $(f_0, f_1, 2f_1 - 2f_0, f_1 - f_0)$.

Recalling that every d dimensional polytope must have $d + 1$ vertices,we start enumerating the possible cases with $f_0 = 5$.

- By the above, we must have $f_1 \geq 10$, and there is guaranteed to be a triangulation if $f_1 \leq 10$. So $(5, 10, 10, 5)$ is one such face vector. By the upper bound of the face vector, this is the only possible face vector.

7

Pre-req: Commutative Algebra 2, Simplicial Complexes

7.1 Stanley-Reisner Rings

Theorem 7.1.1
Let R be a commutative ring. Let $n \in \mathbb{N}$. There is a bijection

$$\begin{matrix}\text{Abstract Simplicial}\\ \text{Complexes on }\{1,\dots,n\}\end{matrix} \overset{1:1}{\longleftrightarrow} \begin{matrix}\text{Squarefree monomial}\\ \text{ideals of }R[x_1,\dots,x_n]\end{matrix}$$

given by sending a simplicial set S to the ideal $\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$.

Definition 7.1.2 (Stanley-Reisner Rings)
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Define the Stanley-Reisner ring of S to be

$$R[S] = \frac{k[x_1,\dots,x_n]}{\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle}$$

The ideal $I_S = \left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$ is called the Stanley-Reisner ideal of S.

Example 7.1.3
Let R be a commutative ring. The Stanley-Reisner ring of the standard n-simplex Δ^n is given by

$$R[\Delta^n] = k[x_1,\dots,x_n]$$

Example 7.1.4
Let R be a commutative ring. The Stanley-Reisner ring of the abstract simplicial complex $S = \mathcal{P}(\{1,2,3\}) \cup \mathcal{P}(\{1,3,4\}) \cup \mathcal{P}(\{2,4\})$ over $\{1,2,3,4\}$ is given by

$$R[S] = \frac{k[x_1,x_2,x_3,x_4]}{(x_1x_2x_4, x_2x_3x_4)}$$

Proposition 7.1.5
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Then we have

$$I_S = \bigcap_{J\in S} \langle x_i \mid i \notin J\rangle$$

Maps of simplicial sets -> Maps of k-algebra.

Proposition 7.1.6
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a flag complex, then $R[\mathcal{K}]$ is a Koszul algebra.

7.2 The Hilbert Series of Stanley-Reisner Rings

Theorem 7.2.1 (Stanley)
Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Let $(f_0,\dots,f_n)$ be the face vector of S. Let $(h_0,\dots,h_n)$ be the h-vector of S. Then the Hilbert series of $k[S]$ is given

by

$$HS_{k[S]}(t) = \sum_{i=0}^{n+1} f_{k-1}\left(\frac{t^2}{1-t^2}\right)^k = \frac{1}{(1-t^2)^{n+1}}\sum_{i=0}^{n+1} h_i t^{2i}$$

Example 7.2.2 | Let Δ^n be the standard n-simplex. Then the following are true.

- The Hilbert series for the Stanley-Reisner ring of Δ^n is given by

$$HS_{k[\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}$$

- The Hilbert series for the Stanley-Reisner ring of $\partial\Delta^n$ is given by

$$HS_{k[\partial\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}\sum_{i=0}^{n+1} t^{2i}$$

8 Moment-Angle Complexes

Pre req: Equivariant (co)homology, homological algebra, PL manifolds?

8.1 Basic Definitions

Proposition 8.1.1
Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \to \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \to (|z_1|^2, \dots, |z_n|^2)$. Suppose that $(S^1)^n \subseteq \mathbb{C}^n$ act on $\mathbb{C}^n$ by left multiplication. Then μ descends to a homeomorphism of orbit spaces

$$\frac{\mathbb{C}^n}{(S^1)^n} \cong \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_i \geq 0 \text{ for all } i\} \subseteq \mathbb{R}^n$$

Lemma 8.1.2
Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \to \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \to (|z_1|^2, \dots, |z_n|^2)$. Then the above homeomorphism induces a homeomorphism

$$\frac{D^n}{(S^1)^n} \cong [0, 1]^n$$

Recall the construction of the cubical embedding $i_c : \Delta^n \to [0, 1]^{n+1}$.

1. For $1 \leq k \leq n$, the barycenter of the k-dimensional face $B([e_{i_1}, \dots, e_{i_k}])$ is sent to $(u_1, \dots, u_n) \in [0, 1]^{n+1}$ where $u_j = 0$ if $j \in \{i_1, \dots, i_k\}$ and $u_j = 1$ otherwise.
2. Extend this into a piecewise linear embedding $i_c : \Delta^n \to [0, 1]^{n+1}$ as follows. For each simplex $[p_1, \dots, p_l]$ in the barycentric subdivision of Δ^n, $[i_c(p_1), \dots, i_c(p_l)]$ lies on a face of $[0, 1]^{n+1}$. Hence there exists an simplicial map sending $[p_1, \dots, p_l]$ to $[i_c(p_1), \dots, i_c(p_l)]$.

Since every finite abstract simplicial complex can be realized as a subcomplex of Δ^n for large n, we also get the cubical embedding of a finite abstract simplicial complex, which is denoted by $\text{cc}(\mathbb{K})$.

Definition 8.1.3 (Moment-Angle Complex)
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Define the moment-angle complex of $\mathcal{K}$ to be

$$\mathcal{Z}_{\mathcal{K}} = \mu^{-1}(\text{cc}(\mathcal{K})) \subseteq D^n$$

Equivalently, it is the pullback of the inclusion map $\mathcal{Z}_{\mathcal{K}} = \text{cc}(\mathcal{K}) \to [0, 1]^n$ and the quotient map $\mu : D^n \to [0, 1]^n$.

Example 8.1.4
Let $n \in \mathbb{N}$. Then the following are true.
- $\mathcal{Z}_{\emptyset} = (S^1)^n$.
- $\mathcal{Z}_{\Delta^n} = D^{n+1}$.

Lemma 8.1.5
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. The following are true regarding the moment-angle complex of $\mathcal{K}$.
- $\mathcal{Z}_{\mathcal{K}}$ is a $(S^1)^n$-invariant subspace.
- The homeomorphism $\frac{D^n}{(S^1)^n} \cong [0, 1]^n$ induced by the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$ descends to a homeomorphism
$$\frac{\mathcal{Z}_{\mathcal{K}}}{(S^1)^n} \cong \text{cc}(\mathcal{K})$$

Recall that $\text{cc}(\mathcal{K})$ is equal to the union of a finite number of faces of $[0, 1]^n$. Also recall the notation

$$C_I = \{(x_1, \dots, x_n) \in [0, 1]^n \mid x_i = 0 \text{ if } i \in I\}$$

for a face of the cube $[0, 1]^n$ that contains the origin and $I \subseteq \{1, \dots, n\}$.

Lemma 8.1.6
Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Suppose that $\text{cc}(\mathcal{K}) = \bigcup_{I \in \mathcal{K}} C_I$. Then there is a homeomorphism

$$\mathcal{Z}_{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} \mu^{-1}(C_I) = \bigcup_{I \in \mathcal{K}} \{(x_1, \dots, x_n) \in D^n \mid |z_j|^2 = 1 \text{ for all } j \in \{1, \dots, n\} \setminus I\}$$

where μ is the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$.

For $I \subseteq \{1, \dots, n\}$, the condition that $|z_j|^2 = 1$ for $j \in \{1, \dots, n\} \setminus I$ means that for those coordinates in D^n, they only consist of the boundary which is S^1. Hence we may think of $\mathcal{Z}_{\mathcal{K}}$ as the union of products

$$\mathcal{Z}_{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} \left(\prod_{i \in I} D^2 \times \prod_{i \in \{1, \dots, n\} \setminus I} S^1 \right)$$

But note that this product is ambiguous on which coordinates should be S^1 and which should be D^2 (because we have a chosen embedding into $\mathbb{C}^n$ although swapping the order of the products give homeomorphic spaces). Moreover, because we are taking the union over all $I \in \mathcal{K}$, it suffices to consider the maximal simplicies in $\mathcal{K}$.

Example 8.1.7
Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are true.

- The moment angle complex of $\partial \Delta^n$ is given by

$$\mathcal{Z}_{\partial \Delta^n} \cong S^{2n+1}$$

- The moment angle complex of the n-gon is given by

Proof
It suffices to consider the maximal simplicies in $\partial \Delta^n$. They are of the form $I \subseteq \{1, \dots, n+1\}$ for $|I| = n$. Hence we have

$$\begin{aligned} \mathcal{Z}_{\partial \Delta^n} &= (D^2 \times \cdots \times D^2 \times S^1) \cup \cdots \cup (S^1 \times D^2 \times \cdots \times D^2) \\ &= \partial (D^2 \times \cdots \times D^2) \\ &\cong \partial D^{2n+2} \\ &\cong S^{2n+1} \end{aligned}$$

For the moment angle complex of the n-gon, we note that the simplicial complex is given by 1-simplicies of the form $\{i, i+1\}$ for $1 \leq i \leq n$ together with $\{1, n+1\}$. Hence

$$\mathcal{Z}_{n\text{-gon}} = (D^2 \times D^2 \times S^1 \times \cdots \times S^1) \cup \cdots \cup (S^1 \times \cdots \times S^1 \times D^2 \times D^2) \cup (D^2 \times S^1 \times \cdots \times S^1 \times D^2)$$

■

Proposition 8.1.8
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1}, then $\mathcal{Z}_{\mathcal{K}}$ is a closed topological manifold of dimension $n + m$.

8.2 Polyhedral Products

Definition 8.2.1 (Polyhedral Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A}) = \{(X_1, A_1), \dots, (X_n, A_n)\}$ be pairs of spaces. Define the polyhedral product of $(\mathbf{X}, \mathbf{A})$ corresponding to $\mathcal{K}$ to be

$$(\mathbf{X}, \mathbf{A})^{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} \left\{ (x_1, \dots, x_n) \in \prod_{i=1}^{n} X_i \mid x_i \in A_i \text{ for } i \notin I \right\} = \bigcup_{I \in \mathcal{K}} \left(\prod_{i \in I} X_i \times \prod_{i \notin I} A_i \right)$$

8.3 Algebraic Invariants of the Moment Angle Complex

Proposition 8.3.1
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- There is an isomorphism of rings
$$H^*((\mathbb{CP}^\infty, *)^{\mathcal{K}}; R) \cong R[\mathcal{K}]$$
- The inclusion $(\mathbb{CP}^\infty, *)^{\mathcal{K}} \hookrightarrow (\mathbb{CP}^\infty, *)^n$ induces the quotient map
$$H^*((\mathbb{CP}^\infty)^m; R) \cong R[v_1, \dots, v_n] \to R[\mathcal{K}] \cong H^*((\mathbb{CP}^\infty, *)^{\mathcal{K}}; R)$$
(where $v_1, \dots, v_n$ are degree 2 elements).
- The inclusion $i : (\mathbb{CP}^\infty, *)^{\mathcal{K}} \hookrightarrow (\mathbb{CP}^\infty, *)^n$ decomposes into a homotopy equivalence and fibration in the following way:
$$\begin{tikzcd} (\mathbb{CP}^\infty, *)^{\mathcal{K}} \arrow[rr, hook] \arrow[dr, "\simeq"'] & & (\mathbb{CP}^\infty, *)^n \\ & ET^m \times_{T^m} \mathcal{Z}_{\mathcal{K}} \arrow[ur, "\text{fib.}"'] & \end{tikzcd}$$
Moreover, the fiber of the fibration is $\mathcal{Z}_{\mathcal{K}}$.
- $\mathcal{Z}_{\mathcal{K}}(BS^1, *) = ET^m \times_{T^m} \mathcal{Z}_{\mathcal{K}}$.
- $H^*(ET^m \times_{T^m} \mathcal{Z}_{\mathcal{K}}) = R[\mathcal{K}]$.

Theorem 8.3.2 ((Davis-Januskiewicz))
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the equivariant cohomology ring of $\mathcal{Z}_{\mathcal{K}}$ is given by

$$H^*_{T^m}(\mathcal{Z}_{\mathcal{K}}; R) \cong R[\mathcal{K}]$$

(natural in $\mathcal{K}$)

Proposition 8.3.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true regarding the homotopy groups of $\mathcal{Z}_{\mathcal{K}}$.

- $\mathcal{Z}_{\mathcal{K}}$ is two-connected.
- $\pi_i(\mathcal{Z}_{\mathcal{K}}) \cong \pi_i((\mathbb{CP}^\infty, *)^{\mathcal{K}})$ for $i \geq 3$.

Consider the decomposition of D^2 into the following CW complex structure:

- One 0-cell $1 \in D^2$.
- One 1-cell S^1.
- One 2-cell D^2.

Let $n \in \mathbb{N}$. This gives the product complex D^{2n} a CW complex structure, whose cells are a product of 1-cells and 2-cells. We parameterize the cells in the following way. For $J, I \subseteq \{1, \dots, n\}$ with $J \cap I = \emptyset$, we denote $\chi(J, I)$ as the $(|J| + |I|)$-cell corresponding to the cell with the factor of a 1-cell in the jth position for $j \in J$, the factor of a 0-cell in the ith position for $i \in I$, and the factor of a 0-cell for $k \notin I \cup J$.

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a union of cells in D^{2n}, and so is a cellular subcomplex of D^{2n}. Namely, for each $I \in \mathcal{K}$, we have that $\chi(J, I) \subseteq \mathcal{Z}_{\mathcal{K}}$ for all $J \subseteq \{1, \dots, n\} \setminus I$.

Q: How does the cellular cochain split into a bicomplex

Proposition 8.3.4
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

-
- There is an isomorphism of graded rings

$$H^*(\mathcal{Z}_{\mathcal{K}}; R) \cong H^*\left(\frac{\Lambda_R[u_1, \dots, u_n] \otimes_R R[\mathcal{K}]}{(v_i^2, u_i v_i \mid 1 \leq i \leq n)}\right)$$

Proposition 8.3.5
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is an isomorphism of graded algebras

$$H^*(\mathcal{Z}_{\mathcal{K}}; R) \cong \operatorname{Tor}^*_{R[v_1, \dots, v_m]}(R[\mathcal{K}], R)$$

(natural in $\mathcal{K}$)

9 Combinatorial Commutative Algebra

9.1

Definition 9.1.1 (Monomial Notation)
Let R be a commutative ring. Let $u = (u_1, \dots, u_n) \in \mathbb{N}^n$. Define

$$x^u = x_1^{u_1} \cdots x_n^{u_n} \in R[x_1, \dots, x_n]$$

9.2 The Semi-group Ring

Definition 9.2.1 (The Semi-Group Ring)
Let S be a monoid. Let R be a ring. Define the semi-group ring of S over R to be the left R-module

$$R[S] = \bigoplus_{s \in S} R \cdot s$$

together with multiplication defined on basis by $t^{s_1} \cdot t^{s_2} = t^{s_1 + s_2}$

Lemma 9.2.2
Let S be a monoid. Let R be a ring. Then the following are true.
- $R[S]$ is a ring.
- If S is commutative, then $R[S]$ is a commutative ring.
- If R and S are commutative, then $R[S]$ is an R-algebra.

9.3 Affine Semi-groups

Definition 9.3.1 (Affine Semi-group)
Let S be a monoid. We say that S is an affine semi-group if there exists an injective monoid homomorphism $S \to \mathbb{N}^d$ for some $d \in \mathbb{N}$.

9.4 Lattice Ideals

Let S be a commutative monoid. Write $G(S)$ the grothendieck completion of S. It is an abelian group.

Definition 9.4.1 (Lattice Ideal)
Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Define the lattice ideal of L in $R[x_1, \dots, x_n]$ to be

$$I_L = R\langle x^u - x^v \mid u, v \in \mathbb{N}^n \text{ and } u - v \in L \rangle$$

Proposition 9.4.2
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then there is a k-algebra isomorphism

$$k[S] \cong \frac{k[x_1, \dots, x_n]}{I_{\ker(\phi)}}$$

Proposition 9.4.3
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then the following are equivalent.
- $I_{\ker(\phi)}$ is prime.
- $k[S]$ is an integral domain.
- $G(S)$ is a free abelian group.
- There exists an injective monoid homomorphism $S \to \mathbb{N}^d$ for some $d \in \mathbb{N}$.

10 Constructing Toric Varieties

10.1 The Standard n-Torus

Definition 10.1.1 (The Standard n-Torus)
Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Define the standard n-torus over k to be the affine group variety

$$\mathbb{G}_m^n = \text{Spec}(k[x, x^{-1}])^n$$

Definition 10.1.2 (Tori)
Let k be a field. Let V be an affine algebraic group. We say that V is a torus if there exists a field extension K/k such that the algebraic group V over K is isomorphic to $\mathbb{G}_m^n = \text{Spec}(K[x, x^{-1}])^n$ for some $n \in \mathbb{N} \setminus \{0\}$.

Definition 10.1.3 (Split Tori)
Let k be a field. Let V be a torus over the field extension K/k. We say that V is split if $K = k$.

If K is separably closed or algebraically closed, then all tori are split.

Recall that the functor $D : \mathbf{FGAb} \to \mathbf{Diag}$ sending a finitely generated abelian group G to $D(G) = \text{Spec}(k[G])$ is an equivalence of categories. Evidently split tori corresponds to the free abelian groups of finite rank under the equivalence.

The following are some more properties of a split torus T:

- T is diagonalizable (in T is a torus if and only if T is smooth connected diagonalizable)
- Every representation of G is diagonalizable.
- If V is a representation of T, then we can decompose V into its eigenspaces:

$$V = \bigoplus_{\chi \in \chi(T)} V_\chi$$

Lemma 10.1.4
Let k be a field. Let T be split torus over k. Then the following are true.
- Let T' be a torus over k. Let $\phi : T \to T'$ be a morphism of group schemes. Then $\text{im}(\phi)$ is a torus.
- Let $H \leq T$ be a subgroup variety of T. If H is irreducible, then H is a torus.
- Let $H \leq T$ be a subgroup of T. Then T/H is a torus.

Definition 10.1.5 (Character Lattices)
Let k be a field. Let T be a torus over k. Define the character lattice of T to be the group of characters

$$\chi(T) = \text{Hom}_{\mathbf{GrpSCh}_k}(T, \mathbb{G}_m)$$

Definition 10.1.6 (One-parameter Subgroups)
Let k be a field. Let T be a torus over k. A one-parameter subgroup of T is a homomorphism of group schemes $\mathbb{G}_m \to T$.

Since χ is the inverse to D, we have the following result.

Lemma 10.1.7
Let k be a field. Let T be a split torus over k. Then the following are true.
- The map

$$\chi(T) \times \text{Hom}_{\mathbf{GrpSch}_k}(\mathbb{G}_m, T) \to \mathbb{Z}$$

defined by $(\chi, \lambda) \to \chi \circ \lambda$ is a perfect pairing.
- A choice of an isomorphism $T \cong \mathbb{G}_m^n$ induces a isomorphisms $\chi(T) \cong \mathbb{Z}^n \cong \text{Hom}_{\mathbf{GrpSch}_k}(\mathbb{G}_m, T)$ such that the above perfect pairing is the standard Euclidean dot product.

Proposition 10.1.8
Let k be a field. The functor

$$\chi(-_{k^{\text{sep}}}) : \mathbf{Tori}_k \to \mathbf{FGFreeAb}(\mathbf{withcontinuousactionof})\text{Gal}(k^{\text{sep}}/k)$$

is an equivalence of categories.

10.2 Constructing Toric Varieties

Definition 10.2.1 (Toric Varieties)
Let X be an irreducible variety over $\mathbb{C}$. We say that X is a toric variety if the following are true.
- $(\mathbb{C}^*)^n$ is a Zariski open subset of X.
- The action of $(\mathbb{C}^*)^n$ on itself extends to X.

We denote the underlying torus of X by T_X.

Definition 10.2.2 (Affine Toric Varieties Associated to Characters)
Let $m \in \mathbb{N}^+$. Let $\mathcal{A} = \{\chi_1, \dots, \chi_m\} \subset \chi(\mathbb{G}_m^n)$ be characters of the standard n-torus. Define the affine toric variety associated to $\mathcal{A}$ to be

$$Y_{\mathcal{A}} = \overline{\text{im}(\Phi_{\mathcal{A}})}$$

where $\Phi_{\mathcal{A}} : \mathbb{G}_m^n \to \mathbb{A}_{\mathbb{C}}^m$ is the map defined by $\Phi_{\mathcal{A}}(t) = (\chi_1(t), \dots, \chi_m(t))$

Proposition 10.2.3
Let $m \in \mathbb{N}^+$. Let $\mathcal{A} \subset \chi(\mathbb{G}_m^n)$ be a finite subset of characters of the standard n-torus. Then the following are true.
- $Y_{\mathcal{A}}$ is an affine toric variety.
- The underlying torus of $Y_{\mathcal{A}}$ has character lattice $\mathbb{Z}\mathcal{A}$.
- $\dim(Y_{\mathcal{A}}) = \text{rank}(\mathbb{Z}\mathcal{A})$

Definition 10.2.4 (Toric Ideals)
Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Let $I_L \subseteq R[x_1, \dots, x_n]$ be the lattice ideal of L. We say that I_L is a toric ideal if I_L is a prime ideal.

Proposition 10.2.5
Let X be an irreducible affine variety over $\mathbb{C}$. Then the following are equivalent.
- X is a affine toric variety.
- There exists a lattice L and a finite subset $\mathcal{A} \subseteq L$ such that $X = Y_{\mathcal{A}}$.
- There exists a toric ideal I of $\mathbb{C}[x_1, \dots, x_n]$ such that $X = \text{Spec}\left(\frac{\mathbb{C}[x_1,\dots,x_n]}{I}\right)$.
- There exists an affine semigroup S such that $X = \text{Spec}(\mathbb{C}[S])$.

Proposition 10.2.6
Let k be an algebraically closed field. Let S be an affine semi-group. Then there is a bijection

$$\{p \in \text{Spec}(k[S]) \mid p \text{ is a closed point }\} \overset{1:1}{\longleftrightarrow} \{\phi : S \to (\mathbb{C}, \cdot) \mid \phi \text{ is a monoid homomorphism }\}$$

given as follows. For each $p \in \text{Spec}(k[S])$ a closed point, p corresponds to the monoid homomorphism $\gamma_p : S \to \mathbb{C}$ defined by $\gamma_p(m) = \chi^m(p)$

Proposition 10.2.7
Let k be an algebraically closed field. Let $X = \text{Spec}(k[S])$ be an affine torus over k. Then the action of the torus T_X on X is induced by the k-algebra homomorphism

$$k[S] \to k[\chi(T_X)] \otimes_k k[S]$$

given by $\gamma_p \mapsto \gamma_p \otimes \gamma_p$ where here we identify the closed points on $k[S]$ by the above prp.

10.3 The Dictionary Between Combinatorics and Toric Varieties

Definition 10.3.1 (The Associated Semigroup of a Cone)
Let $C \subseteq \mathbb{R}^n$ be a rational cone. Define the associated semigroup of C to be

$$S_C = C^\vee \cap (\mathbb{Z}^n)^* \subseteq (\mathbb{R}^n)^*$$

Lemma 10.3.2 (Gordon's Lemma)
Let $C \subseteq \mathbb{R}^n$ be a rational polyhedral cone. Then S_C is finitely generated and hence is an affine semigroup.

Definition 10.3.3 (Affine Toric Variety Associated to Rational Cone)
Let $\sigma \subseteq \mathbb{R}^n$ be a rational polyhedral cone. Define the affine toric variety associated to σ to be

$$V_\sigma = \text{Spec}(\mathbb{C}[S_\sigma])$$

Proposition 10.3.4
There is a one-to-one correspondence

$$\{X \text{ is an affine toric variety over } \mathbb{C}\} \stackrel{1:1}{\longleftrightarrow} \{C \text{ is a rational polyhedral cone }\}$$

given by sending a rational polyhedral cone $\sigma \subseteq \mathbb{R}^n$ to $\text{Spec}(\mathbb{C}[S_\sigma])$.

Proposition 10.3.5
Let $\sigma \subseteq \mathbb{R}^n$ be a rational polyhedral cone. Then the following are equivalent.
- $\text{Spec}(\mathbb{C}[S_\sigma])$ is normal.
- S_σ is a saturated affine semi-group.
- σ is strongly convex.
- $\dim(\text{Spec}(\mathbb{C}[S_\sigma]) = n$.

Proposition 10.3.6
Let $X = \text{Spec}(\mathbb{C}[S])$ be an affine toric variety over $\mathbb{C}$ where S is an affine semigroup. Then the normalization of X is given by

$$\tilde{X} = \text{Spec}(\mathbb{C}[S_{\text{Cone}(S)^\vee}])$$

together with the normalization map given by the inclusion $\mathbb{C}[S] \hookrightarrow \mathbb{C}[S_{\text{Cone}(S)^\vee}]$.

Proposition 10.3.7
Let $\sigma \subseteq \mathbb{R}^n$ be a strongly convex rational polyhedral cone. Then $\text{Spec}(\mathbb{C}[S_\sigma])$ is smooth if and only if σ is regular.

Thus we have constructed a dictionary

$$\begin{aligned} \text{Affine Toric Varieties} &\stackrel{1:1}{\longleftrightarrow} \text{Rational Polyhedral Cones} \\ \text{Affine Normal Toric Varieties} &\stackrel{1:1}{\longleftrightarrow} \text{Strongly Convex Rational Polyhedral Cones} \\ \text{Affine Smooth Toric Varieties} &\stackrel{1:1}{\longleftrightarrow} \text{Regular Rational Polyhedral Cones} \end{aligned}$$

10.4 Morphisms between Affine Toric Varieties

10.5 Relation of Toric Varieties and Toric Topology